

Off-school Advanced Learning Programme
Summative Test 2
Applied Probability and Financial Mathematics

Time allowed: 90 minutes

Total: 100 marks

校外進階學習計劃
總結測驗二
應用概率與金融數學
作答時間：90 分鐘
總分：100 分

Instructions.

- Answer all questions. Each question is split into part (a) and part (b).
- Each part has exactly one best answer.
- You may use a calculator.
- Unless otherwise stated, interest rates are annual.
- In financial mathematics questions, clearly identify the valuation date before setting up the equation.
- In probability questions, clearly identify the random variable, distribution, or conditioning event before calculating.
- **Relevant probability and interest rate factor tables are provided in the Appendix at the end of this test paper to simplify your calculations.**

繁體中文翻譯： 作答指示。

- 請作答所有題目。每題均分 (a) 及 (b) 兩部分。
- 每一部分均只有一個最佳答案。
- 可以使用計算機。
- 除非另有說明，利率均指年利率。
- 在金融數學題目中，建立方程前須清楚指出估值日期。
- 在概率題目中，計算前須清楚指出隨機變數、分布或條件事件。
- 方便計算，本試卷末附提供相關概率表及利率因子表。

Part A. Financial Mathematics

繁體中文翻譯： 甲部：金融數學 A school district pays 102,590 today to install a new solar panel array. This green energy investment is expected to generate electricity savings of 60,000 at the end of year 3 and another 60,000 at the end of year 4. The annual effective interest rate for the district's funding is 5%. **繁體中文翻譯：** 某學區今日支付 102,590 以安裝一套新的太陽能板陣列。這項☐色能源投資預期可在第 3 年年末節省 60,000 的電費，☐在第 4 年年末再節省 60,000。該學區資金的年實際利率☐ 5%。

1. (a) Calculate the present value of the cash inflows (the electricity savings at year 3 and year 4) today. [2]

繁體中文翻譯： 計算今日的現金流入現值，即第 3 年及第 4 年的電費節省之現值。

- A. 95,400
B. 101,192
 C. 108,300
 D. 114,286
 E. 120,000

Solution: Teaching Note: *Estimated Time: 1 minute. Difficulty: Very Easy. Scaffolds the concept that NPV is $PV(\text{inflows}) - PV(\text{outflows})$.*

中文解☐： 教學重點：預計需時 1 分鐘。難度：非常容易。此題鋪墊☐現值等於現金流入現值☐現金流出現值的概念。

The present value of the cash inflows, using exact calculations, is:

$$PV = \frac{60,000}{1.05^3} + \frac{60,000}{1.05^4} \approx 51,830.26 + 49,362.15 = 101,192.41.$$

(Note: Using the rounded factor table in the Appendix yields $60,000(0.8638 + 0.8227) = 101,190$, which cleanly points to the same best choice.) The closest answer is 101,192.

中文解☐： 使用精確計算，現金流入的現值☐

$$PV = \frac{60,000}{1.05^3} + \frac{60,000}{1.05^4} \approx 51,830.26 + 49,362.15 = 101,192.41.$$

☐：若使用附☐中的四舍五入因子表，則得到 $60,000(0.8638 + 0.8227) = 101,190$ ，仍明確指向同一最佳選項。因此最接近的答案☐ 101,192。

- (b) Calculate the net present value of this solar panel investment. [3]

繁體中文翻譯： 計算這項太陽能板投資的☐現值。

- A. -1,398**
 B. -699
 C. 699
 D. 1,398
 E. 2,629

Solution: Teaching Note: *Estimated Time: 1 minute. Difficulty: Very Easy.*

中文解題：教學重點：預計需時 1 分鐘。難度：非常容易。

The net present value is the present value of the cash inflows minus the initial cost:

$$\begin{aligned} \text{NPV} &= \text{PV} - 102,590 \\ &\approx 101,192.41 - 102,590 = -1,397.59. \end{aligned}$$

Hence, the answer is $\boxed{-1,398}$.

中文解題：現值等於現金流入現值減初始成本：

$$\begin{aligned} \text{NPV} &= \text{PV} - 102,590 \\ &\approx 101,192.41 - 102,590 = -1,397.59. \end{aligned}$$

因此答案 $\boxed{-1,398}$ 。

2. A family uses “Education Savings Bonds” (which act as zero-coupon bonds) to plan for future university tuition. The prices of three of these bonds are shown below.

繁體中文翻譯：某家庭使用「教育儲蓄債券」（其作用相當於零息債券）為未來大學學費作規劃。下表列出三種此類債券的價格。

Maturity in years	1	2	3
Price per 1 payable at maturity	0.95420	0.90703	0.85892

到期年期	1	2	3
到期支付 1 的價格	0.95420	0.90703	0.85892

- (a) Calculate the implied accumulation factor for a 1-year investment beginning at the end of year 2 and ending at the end of year 3. [2]

Hint: How can we use the prices of the 2-year and 3-year bonds to figure out the multiplier specifically for that third year?

繁體中文翻譯：計算從第 2 年年末開始、到第 3 年年末結束的一年期投資之隱含累積因子。

提示：我們如何利用 2 年期及 3 年期債券的價格，推算專屬於第 3 年的增長倍數？

- A. 1.0450
- B. 1.0505
- C. 1.0560
- D. 1.0615
- E. 1.0670

Solution: Teaching Note: *Estimated Time: 1.5 minutes. Difficulty: Easy. Scaffolds the forward rate relationship without using the term.*

中文解：教學重點：預計需時 1.5 分鐘。難度：容易。此題在不使用專業術語的情下鋪墊遠期利率關。

The accumulation factor from time 2 to time 3 is the ratio of the 2-year bond price to the 3-year bond price:

$$\frac{P(0, 2)}{P(0, 3)} = \frac{0.90703}{0.85892} \approx 1.0560.$$

The correct answer is 1.0560.

中文解：由時間 2 到時間 3 的累積因子 2 年期債券價格與 3 年期債券價格之比：

$$\frac{P(0, 2)}{P(0, 3)} = \frac{0.90703}{0.85892} \approx 1.0560.$$

正確答案 1.0560。

- (b) Calculate the implied annual interest rate earned specifically during the third year. [3]

Hint: If the accumulation factor from part (a) represents growing your money by a certain percentage, what is that percentage rate?

繁體中文翻譯：計算專屬於第 3 年所賺取的隱含年利率。

提示：若 (a) 部分的累積因子代表資金增加了某一百分比，該百分比利率是多少？

- A. 4.8%
- B. 5.0%
- C. 5.2%
- D. 5.4%
- E. 5.6%**

Solution: Teaching Note: *Estimated Time: 1 minute. Difficulty: Easy.*

中文解：教學重點：預計需時 1 分鐘。難度：容易。

The implied interest rate for year 3, let's call it r , is related to the accumulation factor by $1 + r = 1.0560$. Therefore, $r \approx 0.0560$. The correct answer is 5.6%.

中文解：設第 3 年的隱含利率 r ，則其與累積因子的關 $1 + r = 1.0560$ 。因此 $r \approx 0.0560$ ，正確答案 5.6%。

3. Two students are saving money for university. Student A deposits 100 into an account at time 0. The account earns interest at a nominal annual rate of i , compounded semiannually. Student B deposits 200 into a different account at time 0. This second account earns simple interest at the same annual rate i . The two students earn the same amount of interest during the last six months of the eighth year.

繁體中文翻譯：兩名學生正在大學儲蓄。學生 A 在時間 0 將 100 存入一個口，該口按名義年利率 i 計息，每半年利一次。學生 B 在時間 0 將 200 存入另一個口，該口按相同年利率 i 計算單利。兩名學生在第 8 年最後六個月所賺取的利息相同。

- (a) Express the interest earned by Student B (simple interest account) during the last six months of the eighth year in terms of i . [2]

繁體中文翻譯：以 i 表示學生 B（單利）在第 8 年最後六個月賺取的利息。

- A. $50i$
- B. $100i$**
- C. $200i$
- D. $400i$
- E. $100i^2$

Solution: Teaching Note: *Estimated Time: 1.5 minutes. Difficulty: Medium. Ensures students correctly model simple interest for a fractional period before setting up the harder equation.*

中文解：教學重點：預計需時 1.5 分鐘。難度：中等。此題確保學生在建立較難方程前，能正確處理非整年期間的單利。

Student B's account earns simple interest. The principal is 200. The interest earned over any half-year period is:

$$200 \cdot i \cdot \frac{1}{2} = 100i.$$

The correct answer is $100i$.

中文解：學生 B 的按單利計息，本金 200。任何半年的利息

$$200 \cdot i \cdot \frac{1}{2} = 100i.$$

正確答案 $100i$ 。

(b) Calculate i .

[3]

Hint: How can you set up an equation equating the simple interest expression from part (a) to the compound interest earned during the final half-year?

繁體中文翻譯：計算 i 。

提示：如何建立方程，把 (a) 部分的單利表達式與最後半年所賺取的利利息相等？

- A. 9.06%
- B. 9.26%
- C. 9.46%**
- D. 9.66%
- E. 9.86%

Solution: Teaching Note: *Estimated Time: 3 minutes. Difficulty: Medium.*

中文解：教學重點：預計需時 3 分鐘。難度：中等。

Student A's account has semiannual compounding. At time 7.5 (start of the last six months of the 8th year), it has accumulated for 15 half-years. Its balance is $100 \left(1 + \frac{i}{2}\right)^{15}$. The interest earned during the next half-year is:

$$100 \left(1 + \frac{i}{2}\right)^{15} \frac{i}{2}.$$

Equating the two half-year interest amounts gives:

$$\begin{aligned}
 100 \left(1 + \frac{i}{2}\right)^{15} \frac{i}{2} &= 100i \\
 \frac{1}{2} \left(1 + \frac{i}{2}\right)^{15} &= 1 \quad (\text{assuming } i > 0) \\
 \left(1 + \frac{i}{2}\right)^{15} &= 2 \\
 1 + \frac{i}{2} &= 2^{1/15}.
 \end{aligned}$$

Therefore $i = 2(2^{1/15} - 1) \approx 0.0946$. The correct answer is 9.46%.

中文解題：學生 A 的存款每半年複利一次。在時間 7.5（即第 8 年最後六個月開始時），已累積 15 個半年期，其結余 $100 \left(1 + \frac{i}{2}\right)^{15}$ 。下一個半年期賺取的利息

$$100 \left(1 + \frac{i}{2}\right)^{15} \frac{i}{2}.$$

令兩個半年期的利息相等，得

$$\begin{aligned}
 100 \left(1 + \frac{i}{2}\right)^{15} \frac{i}{2} &= 100i \\
 \frac{1}{2} \left(1 + \frac{i}{2}\right)^{15} &= 1 \quad (\text{假設 } i > 0) \\
 \left(1 + \frac{i}{2}\right)^{15} &= 2 \\
 1 + \frac{i}{2} &= 2^{1/15}.
 \end{aligned}$$

因此 $i = 2(2^{1/15} - 1) \approx 0.0946$ 。正確答案 9.46%。

4. A community library needs to pay software subscription fees (liabilities) of 1,000 in six months and another 1,000 in one year. The library uses its reserve funds to buy two types of government bonds to ensure the money is available exactly when needed:

繁體中文翻譯：某社區圖書館需要在六個月後支付 1,000 的軟件訂閱費（負債），在一年後再支付 1,000。圖書館使用儲備金購買兩類政府債券，以確保資金能在需要時準時到位：

Bond	Maturity	Face amount	Coupon information
I	6 months	1,000	8% nominal annual coupon, semiannual
II	1 year	1,000	5% nominal annual coupon, semiannual

債券	到期日	面值	票息資料
I	6 個月	1,000	8% 名義年票息，每半年支付
II	1 年	1,000	5% 名義年票息，每半年支付

- (a) Calculate the number of units of Bond II needed to exactly match the software fee due at the end of year 1.

繁體中文翻譯：計算需要多少單位的債券 II，才能剛好配合第 1 年年末到期的軟件費用。

- A. 0.95000
- B. 0.96250
- C. 0.97561**
- D. 1.00000
- E. 1.02500

Solution: Teaching Note: *Estimated Time: 2 minutes. Difficulty: Medium. Guides the student to work backwards from the longest maturity liability.*

中文解題：教學重點：預計需時 2 分鐘。難度：中等。引導學生由最長到期負債開始倒推。

Bond II pays a coupon plus principal in one year. Its coupon per half-year is $1000(0.05/2) = 25$. Thus, Bond II pays $1000 + 25 = 1025$ at one year. Let y be the units of Bond II.

$$1025y = 1000 \implies y = \frac{1000}{1025} \approx 0.97561.$$

The correct answer is 0.97561.

中文解題：債券 II 在一年後支付票息加本金。其每半年票息 $1000(0.05/2) = 25$ 。因此債券 II 在一年後支付 $1000 + 25 = 1025$ 。設需要 y 單位債券 II，則

$$1025y = 1000 \implies y = \frac{1000}{1025} \approx 0.97561.$$

正確答案 0.97561。

- (b) Calculate the number of units of Bond I needed.

[3]

Hint: How can you set up an equation for the cash flows at 6 months, keeping in mind that Bond II also pays a coupon at that time?

繁體中文翻譯：計算需要多少單位的債券 I。

提示：考慮債券 II 也會在該時點支付票息，如何 F 六個月時的現金流建立方程？

- A. 0.93809**
- B. 0.94293
- C. 0.97561
- D. 0.98345
- E. 1.00000

Solution: Teaching Note: *Estimated Time: 3 minutes. Difficulty: Medium.*

中文解題：教學重點：預計需時 3 分鐘。難度：中等。

Bond I pays its semiannual coupon plus principal in six months: $1000 + 1000(0.08/2) = 1040$. Bond II also pays a coupon of 25 in six months. Let x be the units of Bond I. The cash flow at six months must equal 1000:

$$1040x + 25y = 1000.$$

Substitute $y = 0.97561$:

$$1040x + 25(0.97561) = 1000$$

$$1040x = 1000 - 24.39025$$

$$x = \frac{975.60975}{1040} \approx 0.93809.$$

The correct answer is 0.93809.

中文解題：債券 I 在六個月後支付半年票息加本金，即 $1000 + 1000(0.08/2) = 1040$ 。債券 II 亦會在六個月後支付 25 的票息。設需要 x 單位債券 I，六個月時的現金流必須等於 1000：

$$1040x + 25y = 1000.$$

代入 $y = 0.97561$:

$$1040x + 25(0.97561) = 1000$$

$$1040x = 1000 - 24.39025$$

$$x = \frac{975.60975}{1040} \approx 0.93809.$$

正確答案 0.93809。

5. You borrow 2,000 from a family member to buy a high-performance laptop for your advanced programming courses. You agree to repay the loan with annual payments at the end of each year for 4 years. From your summer job, your first two payments are 500 each. The final two payments are X each. The annual effective interest rate is 5%.

繁體中文翻譯：你向家人借入 2,000，以購買一部用於進階編程課程的高性能手提電腦。你同意在 4 年內每年年末還款。你利用暑期工作的收入，首兩期每期還款 500。最後兩期每期還款 X 。年實際利率 5%。

- (a) Calculate the present value of your first two payments exactly at the time the loan is issued. [2]

Hint: How can we calculate the present value of a regular series of level payments using a standard formula?

繁體中文翻譯：計算首兩期還款在貸款發放時的現值。

提示：如何使用標準公式計算一系列定額付款的現值？

A. 900

B. 930

C. 950

D. 980

E. 1000

Solution: Teaching Note: *Estimated Time: 1.5 minutes. Difficulty: Easy. Scaffolds the setup to isolate the remaining principal required for the second tier of payments.*

中文解E: 教學重點: 預計需時 1.5 分鐘。難度: 容易。此題鋪墊如何分離第二層級付款所需覆蓋的剩餘本金。

The present value of the first 2 payments of 500, evaluated at time 0 at an interest rate of 5%, is:

$$PV = \frac{500}{1.05} + \frac{500}{1.05^2}$$

$$PV \approx 500(0.9524) + 500(0.9070) = 476.20 + 453.50 = 929.70$$

The closest answer is **930**.

中文解E: 首兩期各 500 的還款, 在時間 0 以 5% 利率折現的現值E

$$PV = \frac{500}{1.05} + \frac{500}{1.05^2}$$

$$PV \approx 500(0.9524) + 500(0.9070) = 476.20 + 453.50 = 929.70.$$

最接近的答案E **930**。

(b) Calculate X .

[3]

Hint: What is the remaining principal to be covered by the final two payments if you subtract your answer in part (a) from the initial loan amount?

繁體中文翻譯: 計算 X 。

提示: 若從初始貸款金額扣除 (a) 部分的答案, 最後兩期付款需要覆蓋多少剩餘本金?

A. 605

B. 620

C. 635

D. 650

E. 665

Solution: Teaching Note: *Estimated Time: 3 minutes. Difficulty: Medium.*

中文解E: 教學重點: 預計需時 3 分鐘。難度: 中等。

The total loan amount is 2,000. The portion of the loan covered by the final two payments is:

$$2000 - 929.70 = 1070.30$$

The present value of the final two payments (at time $t = 3$ and $t = 4$) evaluated at time 0 is:

$$\frac{X}{1.05^3} + \frac{X}{1.05^4} = 1070.30$$

Using the interest rate factor table:

$$X(0.8638 + 0.8227) = 1070.30$$

$$X(1.6865) = 1070.30 \implies X = \frac{1070.30}{1.6865} \approx 634.63$$

The closest answer is **635**.

中文解：貸款總額 2,000。由最後兩期付款覆蓋的貸款部分

$$2000 - 929.70 = 1070.30.$$

最後兩期付款（在 $t = 3$ 及 $t = 4$ ）於時間 0 的現值

$$\frac{X}{1.05^3} + \frac{X}{1.05^4} = 1070.30.$$

使用利率因子表：

$$X(0.8638 + 0.8227) = 1070.30,$$

$$X(1.6865) = 1070.30 \implies X = \frac{1070.30}{1.6865} \approx 634.63.$$

最接近的答案 635。

Part B. Probability

繁體中文翻譯：乙部：概率

6. Two expensive 3D printers are purchased for a school engineering lab. An extended warranty (which acts like term insurance) is purchased for each machine. Each warranty has a one-time premium of 500 and provides a 10,000 replacement benefit if that specific printer suffers a complete mechanical failure (breaks down) within 10 years. The probabilities are:

繁體中文翻譯：某學校工程實驗室購買了兩部昂貴的 3D 打印機。每部機器均購買一份延長保養（其作用類似定期保險）。每份保養的一次性保費 500；若該打印機在 10 年內出現完全機械故障（損壞），則提供 10,000 的更換賠償。概率如下：

Outcome at the end of 10 years	Probability
Both printers survive (no failure)	0.960
Only Printer 1 survives	0.025
Only Printer 2 survives	0.010
Neither printer survives	0.005

10 年末的結果	概率
兩部打印機均仍可使用（無故障）	0.960
只有打印機 1 仍可使用	0.025
只有打印機 2 仍可使用	0.010
兩部打印機均不能使用	0.005

- (a) Calculate the expected total claim amount to be paid by the warranty company at the end of 10 years. [2]

Hint: What is the payout for each of the four possible scenarios, and what happens to the payout if neither printer survives?

繁體中文翻譯：計算保養公司在 10 年末預期需要支付的總索償金額。

提示：四種可能情況下各自的賠付是多少？若兩部打印機均不能使用，賠付會怎樣？

- A. 350
- B. 450**
- C. 550
- D. 650
- E. 750

Solution: Teaching Note: *Estimated Time: 1.5 minutes. Difficulty: Very Easy. Scaffolds calculating expected value across discrete states.*

中文解題：教學重點：預計需時 1.5 分鐘。難度：非常容易。此題鋪墊如何在離散狀態下計算期望值。

Expected claim calculation:

- Both survive: claim = 0.
- Only Printer 1 survives (Printer 2 fails): claim = 10,000.

- Only Printer 2 survives (Printer 1 fails): claim = 10,000.
- Neither survives (both fail): claim = 20,000.

$$E[\text{claims}] = 0(0.960) + 10000(0.025) + 10000(0.010) + 20000(0.005) \\ = 250 + 100 + 100 = 450.$$

The correct answer is 450.

中文解：預期索償計算如下：兩部均可使用時，索償 0；只有打印機 1 可使用（打印機 2 故障）時，索償 10,000；只有打印機 2 可使用（打印機 1 故障）時，索償 10,000；兩部均不能使用時，索償 20,000。因此

$$E[\text{claims}] = 0(0.960) + 10000(0.025) + 10000(0.010) + 20000(0.005) \\ = 250 + 100 + 100 = 450.$$

正確答案 450。

(b) Calculate the expected excess of premiums over claims.

[3]

繁體中文翻譯：計算保費超出索償的期望金額。

- A. 450
- B. 550**
- C. 650
- D. 750
- E. 850

Solution: Teaching Note: *Estimated Time: 1.5 minutes. Difficulty: Very Easy.*

中文解：教學重點：預計需時 1.5 分鐘。難度：非常容易。

The total premium collected is $500 + 500 = 1,000$. The expected excess is the total premiums minus the expected claims:

$$1000 - 450 = 550.$$

The correct answer is 550.

中文解：總保費收入 $500 + 500 = 1,000$ 。預期超額等於總保費減預期索償：

$$1000 - 450 = 550.$$

正確答案 550。

7. A local school board runs a device insurance program for student laptops. An actuary studies 100 independent accidental damage claims. The claim repair sizes have a mean of 1,000 and a standard deviation of 400.

繁體中文翻譯：某地區學校董事會學生手提電腦營運一項設備保險計劃。一名精算師研究 100 宗相互獨立的意外損壞索償。每宗維修索償金額的平均數 1,000，標準差 400。

- (a) Calculate the standard deviation of the total claim amount for the 100 claims. [2]

Hint: When adding up independent claims, what statistical measure of spread can be added together directly, and how do we convert it back to standard deviation at the end?

繁體中文翻譯：計算這 100 宗索償總金額的標準差。

提示：把獨立索償相加時，哪一個衡量分散程度的統計量可以直接相加？最後如何轉回標準差？

- A. 400
- B. 1,000
- C. 4,000**
- D. 10,000
- E. 40,000

Solution: Teaching Note: *Estimated Time: 1.5 minutes. Difficulty: Easy. Helps avoid the common trap of multiplying the standard deviation by n instead of $\sqrt{n}$.*

中文解題：教學重點：預計需時 1.5 分鐘。難度：容易。此題協助避免常見錯誤，即把標準差乘以 n 而非乘以 $\sqrt{n}$ 。

Let $S = X_1 + X_2 + \cdots + X_{100}$. Since the claims are independent:

$$\text{Var}(S) = 100 \times \text{Var}(X) = 100 \times (400)^2.$$

$$\text{SD}(S) = \sqrt{100} \times 400 = 10 \times 400 = 4000.$$

The correct answer is 4,000.

中文解題：設 $S = X_1 + X_2 + \cdots + X_{100}$ 。由於索償彼此獨立，

$$\text{Var}(S) = 100 \times \text{Var}(X) = 100 \times (400)^2.$$

因此

$$\text{SD}(S) = \sqrt{100} \times 400 = 10 \times 400 = 4000.$$

正確答案 4,000。

- (b) Using a normal approximation, calculate the probability that the total claim amount is less than 92,000. [3]

Hint: How can you apply the Central Limit Theorem to find the Z-score?

繁體中文翻譯：使用常態近似，計算總索償金額少於 92,000 的概率。

提示：如何應用中央極限定理求出 Z 分數？

- A. 0.023**
- B. 0.050
- C. 0.421
- D. 0.579
- E. 0.977

Solution: Teaching Note: *Estimated Time: 2 minutes. Difficulty: Easy.*

中文解題：教學重點：預計需時 2 分鐘。難度：容易。

$E[S] = 100 \times 1000 = 100,000$. By the central limit theorem, $S \approx N(100000, 4000^2)$. Standardize:

$$Z = \frac{S - 100000}{4000}.$$

$$P(S < 92000) = P\left(Z < \frac{92000 - 100000}{4000}\right) = P(Z < -2).$$

From the standard normal table (Table 1), $P(Z < -2) = 0.0228$. The closest answer is **0.023**.

中文解： $E[S] = 100 \times 1000 = 100,000$ 。根據中央極限定理， $S \approx N(100000, 4000^2)$ 。標準化得

$$Z = \frac{S - 100000}{4000}.$$

因此

$$P(S < 92000) = P\left(Z < \frac{92000 - 100000}{4000}\right) = P(Z < -2).$$

由標準常態表（表 1）得 $P(Z < -2) = 0.0228$ 。最接近的答案是 **0.023**。

8. A family health insurance plan offers three optional wellness add-ons: Dental, Vision, and Physical Therapy. Each enrolled employee either chooses exactly two of the three add-ons or chooses no add-ons at all. The proportions of employees choosing Dental, Vision, and Physical Therapy are respectively $\frac{1}{4}$, $\frac{1}{3}$, $\frac{5}{12}$.

繁體中文翻譯：某家庭健康保險計劃提供三項可選健康附加保障：牙科、視力及物理治療。每名參與員要剛好選擇三項中的兩項附加保障，要完全不選擇任何附加保障。選擇牙科、視力及物理治療的員比例分別 $\frac{1}{4}$, $\frac{1}{3}$, $\frac{5}{12}$ 。

- (a) Calculate the proportion of employees who choose exactly two wellness add-ons. [2]

Hint: Imagine a group of 120 employees. If we simply add together the number of people choosing each of the three add-ons, what happens?

繁體中文翻譯：計算剛好選擇兩項健康附加保障的員比例。

提示：想像有 120 名員。若我們把選擇三項附加保障的人數直接相加，會發生甚麼？

- A. $1/4$
- B. $1/3$
- C. $5/12$
- D. $1/2$**
- E. $7/12$

Solution: Teaching Note: *Estimated Time: 2.5 minutes. Difficulty: Medium. Isolates the key logical step of understanding that summing the marginal proportions double-counts the dual-coverage group.*

中文解：教學重點：預計需時 2.5 分鐘。難度：中等。此題突顯關鍵邏輯：把邊際比例相加，會把選擇兩項保障的人重計算。

Let p be the proportion of employees who choose exactly two supplementary coverages. Each of these employees contributes 2 to the total count of

coverage selections. The sum of the individual coverage proportions is:

$$\frac{1}{4} + \frac{1}{3} + \frac{5}{12} = \frac{3}{12} + \frac{4}{12} + \frac{5}{12} = 1.$$

Therefore, $2p = 1 \implies p = 1/2$. The correct answer is $\boxed{1/2}$.

中文解：設剛好選擇兩項附加保障的員工比例 p 。每名此類員工會在保障選擇總數中貢獻 2 次。各項保障的比例總和

$$\frac{1}{4} + \frac{1}{3} + \frac{5}{12} = \frac{3}{12} + \frac{4}{12} + \frac{5}{12} = 1.$$

因此 $2p = 1 \implies p = 1/2$ 。正確答案 $\boxed{1/2}$ 。

- (b) Calculate the probability that a randomly selected employee chooses no wellness add-ons. [3]

Hint: Considering every employee chooses either exactly two add-ons or none at all, how can you use this nature?

繁體中文翻譯：計算隨機抽取一名員工時，該員工不選擇任何健康附加保障的概率。

提示：既然每名員工要剛好選擇兩項附加保障，要完全不選，如何利用這一特性？

- A. 0
- B. $\frac{47}{144}$
- C. $\frac{1}{2}$
- D. $\frac{97}{144}$
- E. $\frac{7}{9}$

Solution: Teaching Note: *Estimated Time: 1.5 minutes. Difficulty: Easy once Part A is solved.*

中文解：教學重點：解 (a) 部分後，本題預計需時 1.5 分鐘。難度：容易。

Employees either choose exactly two coverages or no coverage. The proportion choosing no supplementary coverage is:

$$1 - p = 1 - \frac{1}{2} = \frac{1}{2}.$$

The correct answer is $\boxed{1/2}$.

中文解：員工要剛好選擇兩項保障，要不選任何保障。因此不選擇任何附加保障的比例

$$1 - p = 1 - \frac{1}{2} = \frac{1}{2}.$$

正確答案 $\boxed{1/2}$ 。

9. A city council establishes a community grant fund of 120 (in thousands). It will pay an award of C thousand to each local start-up business that achieves a strict “Green Energy Certification” during the coming year. There are 20 start-ups applying. Each start-up independently has a 0.02 probability of achieving the certification. Let N be the random number of start-ups that successfully certify.

繁體中文翻譯：某市議會設立一項社區資助基金，金額£ 120（以千元£單位）。在未來一年£，每間取得嚴格「£色能源認證」的本地初創企業，均會獲發 C 千元£金。共有 20 間初創企業申請。每間初創企業獨立取得認證的概率£ 0.02。設 N £成功取得認證的初創企業數目。

- (a) Calculate the probability that 3 or more start-ups achieve the certification.

[2]

繁體中文翻譯：計算有 3 間或以上初創企業取得認證的概率。

A. 0.0006

B. 0.0071

C. 0.0599

D. 0.2725

E. 0.3324

Solution: Teaching Note: *Estimated Time: 2 minutes. Difficulty: Easy. Requires finding a simple tail probability from the provided table to setup part (b).*

中文解£：教學重點：預計需時 2 分鐘。難度：容易。此題需要由提供的表格找出簡單尾端概率，£ (b) 部分作準備。

Let $N \sim \text{Binomial}(20, 0.02)$. We want to find $P(N \geq 3)$.

$$\begin{aligned} P(N \geq 3) &= 1 - P(N = 0) - P(N = 1) - P(N = 2) \\ &= 1 - (0.6676 + 0.2725 + 0.0528) \\ &= 1 - 0.9929 = 0.0071 \end{aligned}$$

The closest answer is 0.0071.

中文解£：令 $N \sim \text{Binomial}(20, 0.02)$ 。我們要求 $P(N \geq 3)$ ：

$$\begin{aligned} P(N \geq 3) &= 1 - P(N = 0) - P(N = 1) - P(N = 2) \\ &= 1 - (0.6676 + 0.2725 + 0.0528) \\ &= 1 - 0.9929 = 0.0071. \end{aligned}$$

最接近的答案£ 0.0071。

- (b) Determine the largest value of C such that the probability of the grant fund running out of money is strictly less than 1%.

[3]

Hint: If the probability of running out must be strictly less than 1%, what is the maximum number of successful start-ups the fund must be prepared to fully pay out?

繁體中文翻譯：求 C 的最大值，使資助基金不£支付的概率嚴格小於 1%。

提示：若資金不足的概率必須嚴格小於 1%，基金最多必須準備足£支付多少間成功初創企業？

A. 40

B. 60

C. 80

D. 100

E. 120

Solution: Teaching Note: *Estimated Time: 3 minutes. Difficulty: Medium.***中文解題：**教學重點：預計需時 3 分鐘。難度：中等。

The probability of the fund being inadequate must be strictly less than 0.01. From part (a), $P(N \geq 3) = 0.0071$, which is already less than 0.01. This means the fund is allowed to be inadequate if 3 or more start-ups qualify, but it must be large enough to cover the payout if exactly 2 start-ups qualify. Therefore, the maximum number of guaranteed covered start-ups is 2.

$$2C \leq 120$$

$$C \leq 60.$$

The largest possible value is $\boxed{60}$.

中文解題：基金不足的概率必須嚴格小於 0.01。由 (a) 部分可知 $P(N \geq 3) = 0.0071$ ，已小於 0.01。這表示若有 3 間或以上初創企業合資格，基金可以不足；但若剛好有 2 間合資格，基金必須足以支付。因此基金最多必須保證覆蓋 2 間初創企業：

$$2C \leq 120$$

$$C \leq 60.$$

最大可能值 $\boxed{60}$ 。

10. A basic smartphone insurance policy reimburses one screen-repair claim per year. Out of a large group of student policyholders, 20% will have no screen breaks in the next year (meaning their claim amount is 0). The other 80% will break their screens, and their repair cost is uniformly distributed between 500 and 1,500. The overall median claim amount m across **all** policies satisfies the condition that exactly 50% of all students have a claim of m or less.

繁體中文翻譯：一份基本智能手機保險每年賠償一次屏幕維修索償。在大量學生投保人中，20% 在未來一年不會摔破屏幕（即索償金額 $\boxed{0}$ ）。其餘 80% 會摔破屏幕，而其維修費用在 500 至 1,500 之間均 $\boxed{U}$ 分布。跨越**所有**保單的整體索償金額中位數 m 滿足以下條件：剛好 50% 的學生索償金額小於或等於 m 。

- (a) Calculate the proportion of students *within the broken-screen group* whose repair cost is m or less. [2]

Hint: Imagine a group of 100 students. How many already have a claim of 0? How many have a claim of m or less? How many of the broken-screen students do you need?

繁體中文翻譯：計算在摔破屏幕的學生組 $\boxed{U}$ $\boxed{U}$ ，維修費用小於或等於 m 的學生比例。提示：想像有 100 名學生。有多少人已經有 0 的索償？需要有多少人的索償小於或等於 m ？需要多少名摔破屏幕的學生符合條件？

- A. 0.250
- B. 0.300
- C. 0.375**
- D. 0.500

E. 0.625

Solution: Teaching Note: *Estimated Time: 3 minutes. Difficulty: Medium.* Bypasses formal conditional probability by using a frequency approach (imagining 100 students), which intuitively prevents the common mistake of answering 0.30.

中文解題：教學重點：預計需時 3 分鐘。難度：中等。此題使用頻率方法（想像 100 名學生）避開形式化條件概率，直觀地避免誤答 0.30 的常見錯誤。

Imagine a population of 100 students:

- 20 students have no breaks (claim = 0). Since $0 \leq m$, these 20 automatically count towards the median group.
- 80 students have broken screens.

We need exactly 50 out of the 100 total students to have a claim $\leq m$. Since we already have 20 from the no-break group, we need exactly $50 - 20 = 30$ more students. These 30 students must come from the 80 who broke their screens. Therefore, the proportion of the broken-screen group is:

$$\frac{30}{80} = 0.375.$$

The correct answer is 0.375.

中文解題：想像共有 100 名學生：20 名學生沒有摔破屏幕（索償 0）。由於 $0 \leq m$ ，這 20 人自動計入中位數以下或相等的組；80 名學生摔破屏幕。整體需要剛好 50 名學生的索償金額 $\leq m$ 。既然已經有 20 名來自無損壞組，還需要 $50 - 20 = 30$ 名學生。這 30 名必須來自 80 名摔破屏幕的學生。因此在摔破屏幕組的比例

$$\frac{30}{80} = 0.375.$$

正確答案 0.375。

(b) Calculate the overall median claim amount m .

[3]

Hint: How can you solve for m by taking the proportion you just found in part (a) and setting up an equation?

繁體中文翻譯：計算整體索償金額中位數 m 。

提示：如何利用 (a) 部分剛求出的比例建立方程，從而求出 m ？

- A. 750
- B. 800
- C. 850
- D. 875**
- E. 900

Solution: Teaching Note: *Estimated Time: 2 minutes. Difficulty: Easy.*

中文解題：教學重點：預計需時 2 分鐘。難度：容易。

From part (a), we know that 37.5% of the broken-screen claims must be m

or less. For a uniform distribution on the interval $[500, 1500]$, the proportion of claims below m is given by:

$$\frac{m - 500}{1500 - 500} = \frac{m - 500}{1000}.$$

Setting this equal to 0.375:

$$\begin{aligned}\frac{m - 500}{1000} &= 0.375 \\ m - 500 &= 375 \\ m &= 875.\end{aligned}$$

The correct answer is $\boxed{875}$.

中文解：由 (a) 部分可知，摔破屏幕索償中必須有 37.5% 小於或等於 m 。對於區間 $[500, 1500]$ 上的均分佈，小於 m 的比例

$$\frac{m - 500}{1500 - 500} = \frac{m - 500}{1000}.$$

令其等於 0.375:

$$\begin{aligned}\frac{m - 500}{1000} &= 0.375 \\ m - 500 &= 375 \\ m &= 875.\end{aligned}$$

正確答案 $\boxed{875}$ 。

Appendix: Selected Tables

繁體中文翻譯： 附F：選定表格

Table 1. Standard Normal Cumulative Probabilities $\Phi(z)$

This table provides values for the cumulative distribution function of the standard normal distribution, $\Phi(z) = P(Z \leq z)$.

繁體中文翻譯：表 1：標準常態累積概率 $\Phi(z)$ 。本表提供標準常態分布的累積分布函數值，即 $\Phi(z) = P(Z \leq z)$ 。

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859

Table 2. Binomial Probabilities $P(N = k)$

This table provides exact values for $P(N = k)$ where $N \sim \text{Binomial}(n = 20, p = 0.02)$.

繁體中文翻譯：表 2：二項概率 $P(N = k)$ 。本表提供當 $N \sim \text{Binomial}(n = 20, p = 0.02)$ 時 $P(N = k)$ 的精確值。

k	$P(N = k)$
0	0.6676
1	0.2725
2	0.0528
3	0.0065
4	0.0006

Table 3. Selected Interest Rate Factors

This table provides pre-calculated accumulation and discount factors for specific interest rates.

繁體中文翻譯：表 3：選定利率因子。本表提供特定利率下預先計算的累積因子及折現因子。

n	1.05^n	1.05^{-n}
1	1.050000	0.952381
2	1.102500	0.907029
3	1.157625	0.863838
4	1.215506	0.822702
5	1.276282	0.783526
6	1.340096	0.746215
7	1.407100	0.710681
8	1.477455	0.676839
9	1.551328	0.644609
10	1.628895	0.613913

Table 4. Selected Interest Rate Factors

This table provides pre-calculated accumulation factors for specific interest rates.

繁體中文翻譯：表 4：選定利率因子。本表提供特定利率下預先計算的累積因子。

i	$(1+i)^8$	$(1+i)^9$	$(1+i)^{10}$	$(1+i)^{11}$	$(1+i)^{12}$	$(1+i)^{13}$	$(1+i)^{14}$	$(1+i)^{15}$	$(1+i)^{16}$
4.53%	1.42537	1.48994	1.55743	1.62799	1.70173	1.77882	1.85940	1.94363	2.03168
4.63%	1.43632	1.50282	1.57240	1.64520	1.72137	1.80107	1.88446	1.97171	2.06300
4.73%	1.44733	1.51579	1.58749	1.66258	1.74122	1.82358	1.90983	2.00017	2.09478
4.83%	1.45843	1.52887	1.60271	1.68012	1.76127	1.84634	1.93552	2.02901	2.12701
4.93%	1.46959	1.54205	1.61807	1.69784	1.78154	1.86937	1.96153	2.05824	2.15971
9.06%	2.00135	2.18268	2.38043	2.59609	2.83130	3.08782	3.36757	3.67267	4.00542
9.26%	2.03090	2.21897	2.42444	2.64895	2.89424	3.16225	3.45507	3.77501	4.12457
9.46%	2.06084	2.25579	2.46919	2.70277	2.95846	3.23833	3.54467	3.88000	4.24705
9.66%	2.09115	2.29316	2.51468	2.75760	3.02398	3.31610	3.63643	3.98771	4.37292
9.86%	2.12186	2.33108	2.56092	2.81343	3.09083	3.39559	3.73039	4.09821	4.50229